

RADHEY KALRA

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Building ML/audio systems and self-hosted infrastructure, with current research focused on **audio denoising**.
Leading cloud initiatives at **GDGoC** and **AWS Club**, and interested in scalable infrastructure and collaborative projects.

EDUCATION

Bachelor of Technology in CSE

Model Institute of Engineering and Technology

Aug. 2023 – May 2027

Jammu and Kashmir, India

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, Go, Rust, SQL, Bash, QML, LaTeX

Frameworks & Platforms: Svelte-Kit, React, Next.js, Astro, Tailwind CSS, React Native, Expo, Bun, Node.js

Developer Tools: Git, Docker, CMake, CI/CD, Cloudflare, Vercel, Kubernetes, Terraform

Databases & Cloud: Convex, PostgreSQL, MySQL, AWS, [GCP](#)

Libraries: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Manim

Domains: Deep Learning, Audio Processing, Linux Servers, Full-Stack Development, Networking

Operating Systems: Linux and Windows

EXPERIENCE

Cloud Lead

GDGoC (Google Developer Groups on Campus), MIET Jammu

Aug. 2025 – Present

Jammu and Kashmir, India

- Led cloud learning initiatives by organizing hands-on workshops and guided sessions around **Google Cloud** services and developer tooling.
- Built and maintained a **tutorial platform** containing lab walkthroughs, notes, and video resources to help students complete Study Jam skill badges efficiently; live at [Website](#) with source on [GitHub](#).
- Used **TypeScript, Docker, Terraform, and Linux-based workflows** to streamline deployment and improve the learning experience for campus participants.

Cloud and DevOps Lead

AWS Cloud Club MIET Jammu

Jan. 2026 – Present

Jammu and Kashmir, India

- Leading cloud and DevOps initiatives as part of the core team by mentoring students through workshops, collaborative projects, and technical sessions.
- Promoted adoption of **AWS**, containerization, and infrastructure best practices across club activities with focus on deployment, automation, and scalability.

PROJECTS

Unet Audio Filter | *Audio Enhancement by removing background noise* | [GitHub](#)

- Current research project on audio enhancement using a **custom U-Net-inspired architecture** and end-to-end training pipeline for denoising and source separation.
- Developing the workflow with **Python, PyTorch, and audio DSP preprocessing**, including custom data handling, spectrogram-based modeling, and evaluation tooling.
- In development, but already outperforming existing **open-source baselines on select metrics**, showing promising near-SOTA results in early experiments.

Homelab File Manager | *Self-hosted file management system for Linux servers* | [Live](#) | [GitHub](#)

- Built a **fast, lightweight** file manager for Linux servers using **Svelte** and **Go**, supporting multi-mount browsing, chunked transfers, and real-time file updates.
- Added support for **most media content** with live preview/streaming, plus **Monaco Editor** integration for viewing and editing text-based files directly in the browser.

ViDown | *Self-hosted yt-dlp web interface for downloading media* | [Live](#) | [GitHub](#)

- Built a self-hosted downloader with **SvelteKit, FastAPI, yt-dlp**, and **Docker**, supporting 1000+ sites, format selection, metadata previews, and real-time download progress.
- Deployed it as part of my homelab workflow to fetch media directly from a browser-based interface while keeping downloads and processing on my own server.

AnyConverter | *Self-hosted file conversion and PDF utility toolkit* | [Live](#) | [GitHub](#)

- Built a self-hosted conversion tool for everyday workflows, supporting **PDF, DOC/DOCX, PPT/PPTX, XLS/XLSX, JPEG, PNG, SVG** and other common formats from a browser-based interface.
- Designed it around privacy-first usage so files stay on my own server instead of being uploaded to third-party services.

Homelab Infrastructure | *Self-hosted server for storage, remote access, and service deployment* | [GitHub](#)

- Built a personal homelab on repurposed hardware running **Ubuntu Server**, using **Docker Compose** for service orchestration, **Cloudflare Tunnel** for secure remote access, and **Traefik** for reverse proxying.
- This is where I host tools such as **AnyConverter**, **Homelab File Manager**, and **ViDown**, and use them directly in my day-to-day workflow.
- Maintained a custom **CI/CD workflow** that tracks GitHub changes, pulls updates onto the server, syncs with self-hosted **Gitea**, and automatically redeploys services.

HowTo | *Minimal technical tutorials and guides site* | [Live](#) | [GitHub](#)

- Built a technical tutorials website using **Astro**, **Tailwind CSS**, and **Bun**, with structured content collections, syntax highlighting, and a clean reading experience.
- Actively publish technical guides across both **hardware** and **software** topics, with practical documentation.

Community Leaderboard | *Dashboard for event tracking, rankings, and community management* | [Live](#) | [GitHub](#)

- Developed a community management dashboard with **Next.js**, **Convex**, and **Clerk** for user profiles, event tracking, points allocation, and leaderboard views.
- Designed admin-focused workflows for managing events, monitoring participation, and maintaining rankings through clean, responsive interfaces.

Hotel Booking | *Full-stack booking platform with auth and admin workflows* | [GitHub](#)

- Built a hotel booking application using **Svelte**, **Node.js**, **Express**, and **MySQL**, featuring authentication, booking management, and admin-facing controls for handling reservations.

Terminal Doom | *Doom-inspired terminal game built with ncurses* | [GitHub](#)

- Created a Doom-styled game in the terminal using **C++**, **ncurses**, and **CMake**, translating arcade-style gameplay and rendering into a text-based interface.

Pico32 | *Serial-only TUI for ESP32 Marauder on generic ESP32 hardware* | [Live](#) | [GitHub](#)

- Developed a lightweight terminal interface for ESP32 Marauder using **C++** and PlatformIO, enabling Wi-Fi and Bluetooth testing workflows over serial on low-cost ESP32 boards.
- Focused on portability and usability by adapting a hardware security toolchain into a compact, keyboard-driven TUI experience.

dots-niri | *Automated Arch Linux dotfiles and installer for the Niri compositor* | [GitHub](#)

- Built a personal **Niri** rice and automation setup using **QML**, **Shell**, **JavaScript**, and **Python**, with an installer that provisions packages, symlinks configs, copies assets, and enables desktop services on Arch Linux.
- Currently my **most popular GitHub project**, with **25 stars**, reflecting community interest in Linux customization and reproducible desktop setup workflows.

WallpyGui | *Wallpaper manager for Wayland compositors* | [GitHub](#)

- Built a wallpaper manager for Wayland-based Linux desktops using **Python** and **GTK4**, simplifying wallpaper selection and management through a clean graphical interface.
- Designed the tool around Linux customization workflows, with emphasis on desktop integration and usability for tiling compositor setups.

iwd-applet | *Tray applet for menu-driven wireless management with iwd* | [GitHub](#)

- Developed a lightweight Linux tray applet in **Python** with **GTK/AppIndicator** to manage Wi-Fi through **iwd**, including network discovery, PSK/open network connection, status details, and auto-refreshing link indicators.

TL;DR AI | *Desktop app to summarize WhatsApp chats using LLMs* | [GitHub](#)

- Built an **Electron** desktop application with **React** and **TypeScript** to summarize WhatsApp group chats using OpenRouter-compatible AI models.
- Users have to login using **QR** and enter their openrouter api key.
- Designed the workflow so users can process their own chat data with flexible model choices, making the app practical for personal productivity use cases.

Tenant Manager | *Mobile app for tracking tenants and electricity usage* | [GitHub](#)

- Built a mobile management app with **React Native**, **Expo**, and **TypeScript** to maintain tenant records and log electricity usage in a simpler field-friendly workflow.

CERTIFICATIONS & PROGRAMS

Google Cybersecurity | [Certificate](#)
Cisco Cybersecurity Operations Fundamentals | [Certificate](#)
IBM SkillsBuild: AI Agent Architect | [Certificates](#)
IDE Bootcamp (AICTE / MoE) | [Certificate](#)

EXTRA-CURRICULARS, LEADERSHIP, & INITIATIVES

Team Leader

Internal Hackathon & Smart India Hackathon (SIH 2024 & 2025) *MIET, Jammu*

- Led multidisciplinary teams for the college’s internal hackathon and Smart India Hackathon selection rounds in both **2024** and **2025**.
- Reached the **top 9 teams in the college** in both years by driving collaborative problem-solving, rapid prototyping, and solution design for real-world challenges.

Team Leader

Feb. 2025

Innovation, Design, and Entrepreneurship (IDE) Bootcamp *IIT Jammu*

- Participated in a 5-day immersive bootcamp organized by IIT Jammu in collaboration with the Wadhwani Foundation, AICTE, and MoE’s Innovation Cell.
- Led a team in developing a “Sports Match Finder” app, applying principles of design thinking and the Lean Canvas model to address market needs.
- Engaged in hands-on activities and real-world case studies, gaining insights into business modeling, product design, and entrepreneurial strategies.